KWENTA

Project Abstract and Native Android Architecture

Beladas, John Harold
Janolino, Justine Rei
Marquez, Bruce Riego

Laguna State Polytechnic University
San Pablo City Campus

Project Abstract

Kwenta is a 2D sari-sari store management game intended for Android that transforms everyday Filipino retail experiences into an accessible, interactive learning experience. Players take the role of Tonyo, a young storekeeper who serves customers, fulfills orders through drag-and-drop, gives sukli, manages product availability, and makes purchasing decisions to expand the store's selection.

The game uses a seven-day narrative campaign to introduce mechanics progressively, including change-making, customer satisfaction, weather-related pressure, and fiesta crowds. Daily goals and end-of-shift summaries provide feedback on the player's performance, while tutorials introduce selected mechanics as they become relevant.

The core workflow is to follow the day's story and goal, prepare the store, serve customers, give change when needed, review the shift summary, and continue the campaign. Supplier purchases are optional and available before the shift opens.

Native Android Architecture & Sensor Stack

Kwenta is built with Godot Engine 4.7.2 Stable and uses GDScript as its primary development language. Its scene-based architecture uses Godot Control nodes for interfaces, Resource files (.tres) for campaign and product data, JSON files for story and store configuration, and GDScript systems for customer orders, shifts, sukli, purchases, goals, dialogue, audio, and progression.

The game targets a landscape-oriented experience. Its project settings define a 1920 × 1080 internal viewport, a 1280 × 720 desktop window override, viewport stretching with the expand aspect setting, the GL Compatibility renderer, and ETC2/ASTC texture import support. Actual device resolution and aspect ratio vary by phone; the desktop window override does not define the Android device's display resolution.

The current interaction model uses touch or mouse-style input for actions such as tapping and dragging. No gameplay use of accelerometer, gyroscope, GPS, camera, or other device sensors was identified in the project scripts reviewed. Android export, installation, and behavior on physical devices still need to be verified as part of deployment testing.